

Smart 230V AC Bulb Holder

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The smart 230V AC bulb holder introduced here is an ultra-simple device with an automatic electric bulb controller switch and a standard electric bulb holder realised with the help of a few inexpensive and easily available components.

The electric bulb plugged into the integrated bulb holder of this fit-and-forget device wakes up at nightfall and ticks over just after the first light of day. The device can be used to drive an ordinary incandescent bulb, a light ballast (for fluorescent and discharge lamps), or a solid-state lighting fixture (CFL and LED bulbs) operating on a current less than or equal to 5A at 230V, 50Hz. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram of the smart bulb holder is shown in Fig. 2. It is built around a bridge rectifier (BR1), timer NE555 (IC2), optocoupler PC817(1), 12V Zener diode (ZD1), rectifier diode 1N4007(D1), 12V single-changeover relay (RL1), and a few other components.

As you can see in the schematic diagram, the day/night sensor is a GL5528 light-dependent resistor (LDR), which forms a simple potential divider with an ordinary 100-kilo-ohm resistor (R2). The variable output voltage from this potential divider is used to control one G5L series (12V/400Ω) electromagnetic relay (RL1) through an NE555 timer (IC2). Note that when the input (pins 2 and 6) of IC2 is above 8V, the output (pin 3) goes low, and when the input is below 4V, the output goes high.

For controlling the bulb, common pole (P) and normally-open (NO) contacts of RL1 are used as a single-pole on/off switch to route AC input supply to the integrated bulb holder. Under normal conditions, RL1 is energised

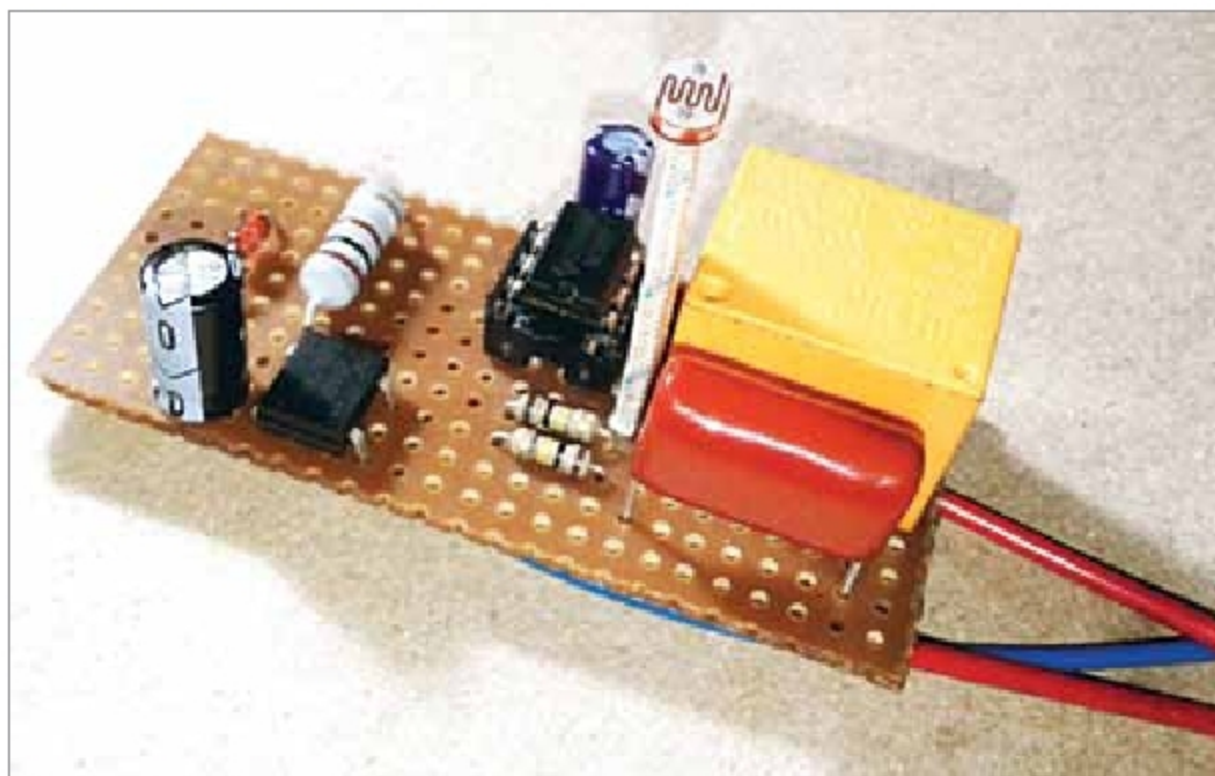


Fig. 1: Author's prototype

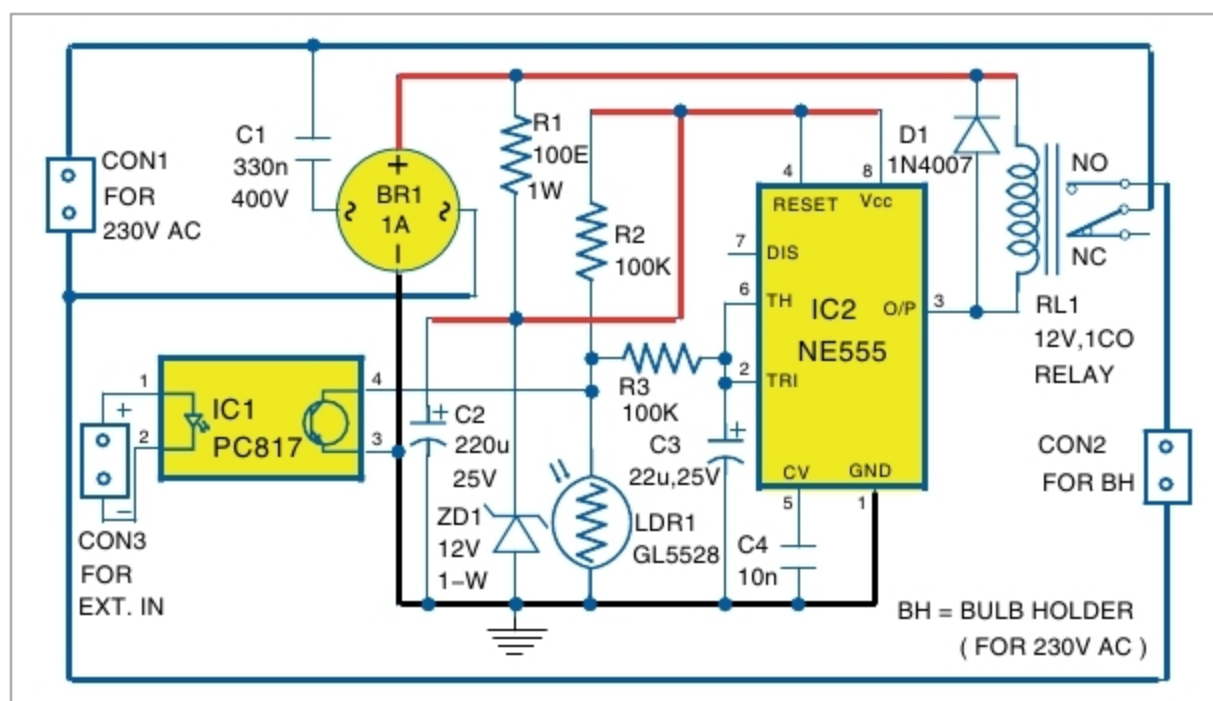


Fig. 2: Circuit diagram of the smart 230V AC bulb holder

through IC1 when ambient light is less than ten lux. When light is greater than forty lux, it will de-energise the relay. You can alter the threshold level of LDR1 by changing the value of the resistor (R2).

The capacitive power supply is used here as the power source. It is configured to cater to a typical output

current of 24mA. IC2 runs on about 12V DC supply available across the 220μF buffer capacitor (C2), while RL1 will get a DC supply of around 14.4V. This 'elevated' 14.4V ensures that RL1 will work smoothly even at the current level that is comparatively much lower than what is actually called for.